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Review

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HER2 expression in breast cancer: evidence gaps and challenges

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The classification of breast cancer (BC) based on HER2 expression is undergoing significant changes. While traditional approaches have focused on HER2-positive and HER2-negative categories, emerging evidence highlights varied therapeutic responses depending on the level of HER2 protein expression. Breast cancers are now immunohistochemically (IHC) scored into five subgroups, which define two primary therapeutic groups: HER2-positive (IHC 2+ amplified and 3 +) and HER2-negative (IHC 0, 1 + , and 2+ non-amplified). Recent advances, particularly in antibody-drug conjugates (ADCs), have led to further subclassification of HER2-negative BC into HER2-Low and HER2-null (IHC 0). Also, for HER-positive subgroups, a differential response to HER2-targeted therapies is seen. This evolving landscape challenges the traditional use of HER2 as a diagnostic marker and underscores the need for a deeper understanding of HER2 biology. This review addresses these complexities, focusing on the emerging HER2-Low and Ultralow subtypes, and evaluates the distinct therapeutic responses across the spectrum of HER2 expression in different BC subtypes.

HER2 protein is an epidermal growth factor receptor (EGFR) of the tyrosine kinase family and forms heterodimers with other ligand-bound EGFR family members such as HER3 and HER4, which activates downstream signalling pathways to promote cell proliferation, cell invasion, angiogenesis and to protect cells against apoptosis (Fig. 1)[1–3]. Normal tissues have a low complement of HER2 membrane protein. Breast cancer (BC) tissue may have up to 25–50 copies of the *HER2* gene and up to 40–100 fold increase in HER2 protein resulting in up to 2 million receptors expressed at the tumour cell surface[4]. Although HER2 overexpression confers a poor prognosis (due to the effect on cell proliferation, migration, invasion, and survival, all hallmarks of cancer), it offers a unique opportunity to utilise oncogenic therapies that target HER2, improving BC patient outcome[5].

In diagnostic pathology, the HER2 status of a breast tumour is assessed using a combination of immunohistochemistry (IHC) to evaluate HER2 protein expression levels, and/or in situ hybridisation (ISH) to assess *HER2* gene status, the only two techniques currently validated for clinical use[6,7]. HER2 positivity in BC is defined as protein overexpression (IHC score 3 +) or HER2 equivocal protein expression (IHC score of 2 +) with evidence of *HER2* gene amplification on ISH reflex testing[6]. This scoring system was developed to predict response to anti-HER2 therapies such as trastuzumab[8] that showed efficacy limited to patients with HER2-positive BC. According to this definition, tumours with equivocal HER2 expression without evidence of *HER2* gene amplification and tumours with low or undetectable levels of HER2 protein (IHC score 1+ or 0, respectively) are defined as HER2 negative to reflect the lack of response to anti-HER2 targeted therapies including trastuzumab and the antibody-drug conjugate (ADC) trastuzumab emtansine (T-DM1)[9]. Recently, the historical HER2 negative group has been sub-classified into 2 subgroups; HER2-Low (IHC score 1 + , and 2 + /ISH-negative) and HER2 0 (IHC score 0) to reflect the difference in potential response to the novel HER2-based ADC therapy (trastuzumab deruxtecan (T-DXd)).

T-DXd, which delivers potent cytotoxic agents to cells with low HER2 levels, has opened a new therapeutic option for this BC patient population and has led to recognition of the concept of HER2-Low BC[9–11]. Fig. 2 provides a chronological outline of the United States Food and Drug Administration (FDA) approvals of ADCs for the treatment of BC.

The spectrum of HER2 expression in BC is evolving. The introduction of the HER2-Low category and the HER2-Ultralow categories has created challenges regarding the significance of the various HER2 categories with regard to potential benefit from adjuvant/neoadjuvant treatment. Furthermore, national guideline recommendations currently vary in their approach to the definition and classification of such new entities[12,13].

Regarding the other end of HER2 expression spectrum, there is also increasing evidence that HER2-positive BCs show differential response rates to anti-HER2 targeted therapies due to various factors, including the level of HER2 protein expression (overexpression (3 +) versus equivocal protein

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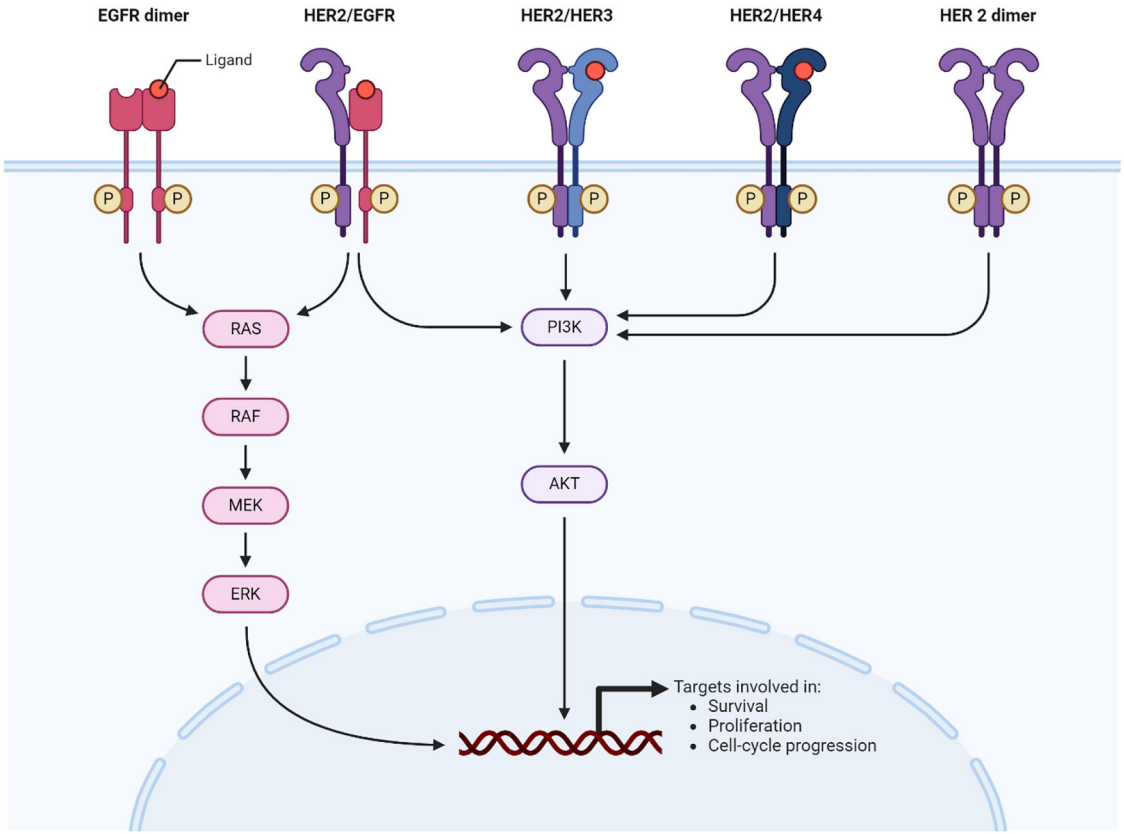

Fig. 1 | HER2 receptor tyrosine kinase signalling pathway. Created with BioRender.com.

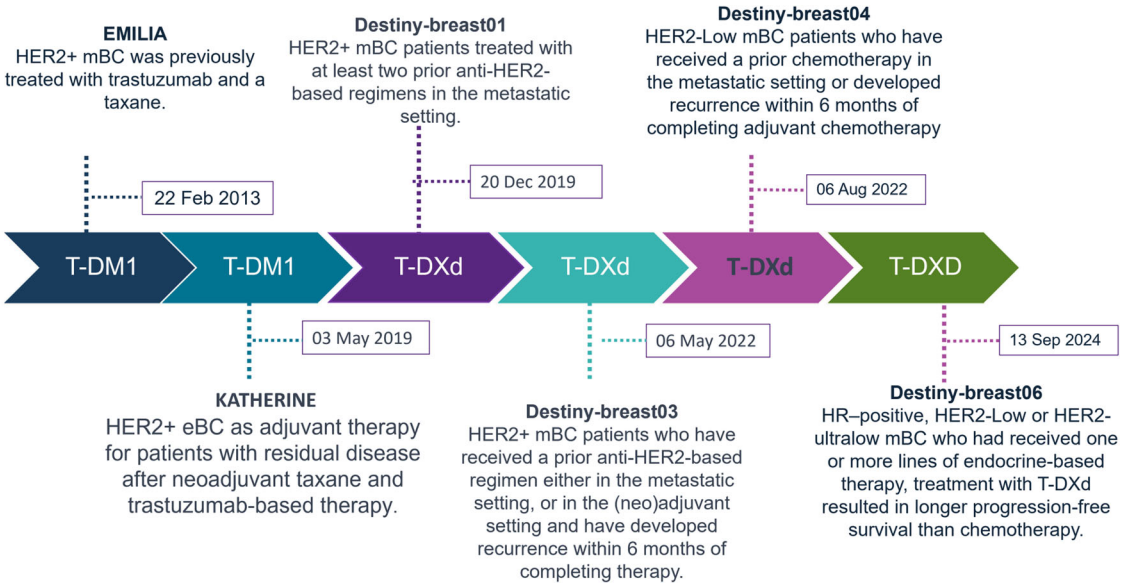

Fig. 2 | Chronological outline of FDA approvals of ADCs for the treatment of breast cancer. ADC: antibody drug conjugate; FDA Food and Drug Administration, HR Hormone receptor, HER2 human epidermal growth factor receptor 2, mBC metastatic breast cancer, pts patients, ChT chemotherapy, T-DM1 trastuzumab emtansine, T-DXd trastuzumab deruxtecan, eBC early breast cancer.

expression (2 +) with *HER2* gene amplification[14], in addition to the effect of concomitant ER expression. This has raised the question as to whether treatment regimes that target HER2 oncogenic activity or ADC-based regimes are more appropriate for patients with HER2 2 + /ISH-positive BC.

These complexities highlight the need for further understanding of the complexities of HER2 biology in BC and the consequent implications for treatment. This review examines the evidence gaps and focuses on HER2-Low and Ultralow tumours and evaluates the different therapeutic responses across the spectrum of HER2 expression in different BC subgroups.

HER2-low BC

Based on the results of the DESTINY-Breast04 trial[15], The FDA approved the use of the HER2-based ADC, T-DXd, for the treatment of HER2-Low

BC. T-DXd showed significant benefit to patients with metastatic HER2-Low BC (mBC), using the current HER2 scoring system (IHC 1+ or 2 + / ISH-negative. Although the HER2-Low BC class has attracted considerable attention in the BC community as a novel therapeutic group of BC, concerns have been raised regarding the validity of the current HER2 companion diagnostic in the identification of these tumours. According to current criteria, approximately 15–20% of BCs are HER2-positive[16–20], 40–60% show lower levels of HER2 protein (HER2-Low) and the remaining BCs (20–45%) lack HER2 expression (IHC score 0), with variation in reported rates in different studies. The HER2-Low category, therefore, comprises a large group of BC patients, approximately 60% of whom have hormone receptor (HR) positive tumours and 40% are HR-negative[21].

Studies aimed at deciphering the biological and clinical significance of HER2-Low BC were carried out to characterise these tumours further[5,22–27], and most of these concluded that HER2-Low BC appears to be a heterogeneous disease[5,28] With regard to its clinical, morphological, immunohistochemical, and molecular characteristics[23,29–32]. Whether HER2-Low BC constitutes a distinct biological entity with specific prognostic and therapeutic implications, possibly influenced by HR status, remains controversial[33–35].

In a previous study, our group concluded that HER2-Low tumours are predominantly of luminal subtype when HR-positive, while if HR-negative, these tumours have a different immunophenotype and are predominantly of luminal 'androgen-like' molecular subtype. In HR-negative BC, HER2-Low BC patients appear to have improved outcomes and are less responsive to neoadjuvant chemotherapy[36].

Irrespective of the ongoing debate regarding the biological significance of HER2-Low phenotype, the clinical relevance of these tumours stems from the presence of HER2 protein on the cell surface of BC cells rather than from the oncogenic activity of the protein. In these tumours, T-DXd[15,37] utilises HER2 as a surface protein to deliver the cytotoxic payload rather than targeting the HER2 oncogene pathway, the mechanism utilised by trastuzumab and related agents[36].

The dilemma of HER2-Ultralow

The response of BC patients with low levels of HER2 protein (including the 1+ category) to T-DXd has led some to expand the definition of BC expressing very low levels of HER2 protein not meeting the IHC 1+ threshold, to a new category called HER2-Ultralow[38]. The oncology and research communities argued that any level of HER2 expression, regardless of its detection using IHC, may be sufficient to elicit an anti-tumour response by T-DXd. One of the major current concerns is the definition of the lowest level of HER2 expression in BC at which T-DXd is effective. In the DESTINY-Breast06 trial, T-DXd was compared with the investigator's choice of chemotherapy in patients with HR-positive mBC, including those with HER2-Ultralow BC ($n = 153$) (defined as HER2 IHC membrane staining but less than score 1 +) and those with HER2-Low BC ($n = 713$)[37]. The study demonstrated that T-DXd significantly improved progression-free survival (PFS) compared to the investigator's choice of chemotherapy in patients with HER2-Low disease (median PFS was 13.2 months for T-DXd ($n = 359$) compared to 8.1 months in the chemotherapy group ($n = 354$; HR, 0.62; 95% CI, 0.51–0.74; $P < 0.0001$). In the intention-to-treat population, which included patients with both HER2-Low and HER2-Ultralow disease, the median PFS was consistent across both treatment arms (HR, 0.63; 95% CI, 0.53–0.75; $P < 0.0001$)[37]. However, the difference in the PFS in the HER2-Ultralow group appeared to be statistically not significant between T-DXd ($n = 76$ patients) and chemotherapy ($n = 76$ patients) (HR, 0.78; 95% CI, 0.50–1.21).

While we appreciate the need to broaden the group of mBC patients who may benefit from T-DXd, and the enthusiasm surrounding the exploration of tumours expressing very low levels of HER2 protein, termed HER2-Ultralow, it is important to recognise the potential disadvantages of this approach. Introducing a new HER2 category at this point may add unnecessary complexity to the identification of such tumours in routine practice. The introduction of a new category to classify HER2-Low/Ultralow tumours, based on any degree of membrane staining, whether complete or incomplete, and present in as few as a single cell up to 9% of tumour cells, may reduce reproducibility and increase interobserver variability. It also raises the question of how many tumour tissue blocks should be examined to confidently determine that a tumour is truly HER2-null. Furthermore, distinguishing between true HER2 staining and technical or staining artefacts presents a significant challenge for pathologists.

If the concept of HER2-Ultralow BC is to be introduced into clinical practice, more evidence from randomised clinical trials is required to confirm that these tumours respond to T-DXd more effectively than completely HER2 IHC 0 tumours (HER2 null). Should data indicate a response of HER2 null tumours to ADC therapy, all patients will be eligible for such treatment without the need for HER2 scoring for selection for this form of treatment.

Another key question is whether HER2 IHC 0 tumours fall within the scope of the T-DXd treatment paradigm. As emerging data begin to shed light on the response of HER2 score 0 BCs, conclusive evidence is still needed to support a meaningful distinction between IHC 1+ and IHC 0 tumours. Such evidence would be essential to justify any revisions to current guideline recommendations[12].

Current challenges in clinical practice

One of the significant challenges that emerged after the results of DESTINY-Breast04 trial is distinguishing HER2 score 0 tumours from those with score 1 + , as the existing American Society of Clinical Oncology/College of American Pathologists (ASCO/CAP) system categorises both as negative, and the scoring system was developed to identify HER2-positive tumours and to identify tumours requiring ISH reflex testing rather than to identify HER2-Low tumours. This makes us question the results of the ongoing clinical trials. In line with this, the DAISY II trial reported a modest level of activity of T-DXd in non-HER2 expressing tumours[38]. However, in the same trial, 48% of the tumours that were HER2 IHC 0 had detectable HER2 expression and were classified as 1+ on external pathology review[38].

The inter-observer agreement among pathologists when distinguishing between HER2 0 and 1+ is relatively poor[39–41]. Also, spatial and temporal heterogeneity in HER2 expression[42] creates challenges in consistently identifying tumours that completely lack membrane staining, increasing the likelihood of false negative results. Furthermore, the sensitivity of current detection kits varies, especially at these very low levels of expression, and significant variation is expected to exist between laboratories using different detection systems. While concordance for HER2-positive and HER2 IHC 0 cases is generally high between central and peripheral laboratories, agreement drops significantly when distinguishing HER2-Low tumours from true negatives. This is mainly due to the subtle and subjective nature of faint or incomplete membrane staining, as well as technical and interpretive variability between laboratories. Central laboratories, which typically follow more rigorous quality assurance protocols and have greater experience, tend to provide more consistent and reproducible HER2-Low scoring compared to local or peripheral centres[13,43,44].

HER2 status is also dynamic in patients with BC, not only in primary versus metastatic sites, but also in repeat biopsies following neoadjuvant therapy[45,46]. These findings will inform our approach to HER2 testing, underscoring the importance of repeat tumour biomarker testing in patients to identify patients who may benefit from ADC treatment.

New testing assay for HER2-Low or refining of the current definition?

The established IHC testing methods (e.g. VENTANA HER2/neu (4B5) assay) are currently used to identify patients with HER2-Low disease in accordance with guideline recommendations[6,15] in many laboratories and are used in the approved and ongoing clinical trials (Tables 1 and 2). The emergence of the HER2-Low and HER2 Ultralow categories highlights the importance of not only adherence to guidelines, but also refining the guideline to improve reproducibility of scoring at the lower end of the spectrum of HER2 expression and the need for further training to ensure

Table 1 | Overview of the clinical trials on ADC in HER2-Low breast cancer

Intervention	Target	Study name	Phase	Patient population	Number	Trial status	Key goal	Key results (or estimated primary completion date)
T-DXd versus physician's choice ChT	HER2	DESTINY-Breast04[15] (randomised, multicentre, two-group, open-label) *NCT03734029*	III	HER2-Low mBC after 1–2 standard therapies	557 (89% HR + , 11% HR −)	Active, not recruiting		**HR+ patients,** cORR: 52.6% versus 16.3% mPFS: 10.1 versus 5.4mo mOS: 23.9 versus 17.5mo **HR− patients,** cORR: 50.0% versus 16.7% mPFS: 8.5 versus 2.9mo mOS: 18.2 versus 8.3mo **Full population**, ILD: 12.1% versus 0.6%
T-DXd versus physician's choice ChT	HER2	DESTINY-Breast06[164] (randomised, multicentre, open-label) *NCT04494425*	III	HR + , HER2-Low (IHC 2 + /ISH-, IHC 1 + , and IHC > 0 0 < 1 +) mBC progressed on ⩾2 lines of ET	866	Complete	PFS	T-DXd significantly improved PFS vs TPC in HER2-low (HR, 0.62 [95% CI 0.51, 0.74], P < 0.0001; median, 13.2 vs 8.1 mo). ITT and HER2-Ultralow results were consistent with HER2-Low
T-DXd	HER2	DEBBRAH[165] (multicentre, open-label, two-stage) *NCT04420598*	II	Pre-treated HER2-positive/HER2-Low unresectable locally advanced or mBC with untreated BMs or LMC (5 cohorts)	41	complete		ORR-ICb: Cohort 2 (HER2± low): 66.7% Cohort 4 (HER2-Low): 33.3% PFS (both cohorts): 5.7mo
T-DXd	HER2	DAISY[166] (open-label) *NCT04132960*	II	HER2-positive, HER2-Low, and HER2-null advanced BC	176 (FAS) (72 HER2-Low)	Active, not recruiting		HER2-Low (HER2-null): BOR: 33.3% (30.6%) mPFS: 6.7mo (4.2mo) ILD: n = 4 (full population)
T-DXd	HER2	Dose-expansion study[167] *NCT02564900*	Ib	HER2-Low advanced/mBC refractory to standard therapies	54 (87% HR + ,13% HR −)	Active, not recruiting		cORR: 37.0%, mPFS: 11.1mo, mOS: 29.4mo, ILD: 14.8%
T-DXd Combinations (Capivasertib, Anastrozole, fulvestrant)	HER2	**DESTINY-Breast08[168] (Multicenter, open label, modular, two-part)** *NCT04556773*	**Ib/II**	HR+ and HR − , HER2-Low advanced or mBC (Part 2: 5 cohorts)	182 (est.)	Active, not recruiting	AEs	Preliminary results, modules 4 and 5, Recommended doses for expansion: T-DXd (5.4 mg/kg) Q3W + Anastrozole (1 mg QD) or Fulvestrant (500 mg Q4Wd) No DLTs reported, Both combinations generally well tolerated
T-DXd+ nivolumab	**HER2**	**Two-part, open-label study[169]** *NCT03523572*	**Ib**	**HER2-expressing mBC after standard therapy**	**Part 2: 45 (16 HER2-Low; 13 HR +)**	**Active, not recruiting**	**HER2-Low:** **ORR: 37.5% mPFS: 6.3mo** **ILD: 10.4% (full population)**	
T-DXd+ durvalumab+ Dato-DXd	HER2	BEGONIA[170] (Two-part, multicentre, multi-arm, open-label) *NCT03742102*	Ib/II	Untreated, unresectable, locally advanced/ metastatic HER2-Low TNBC (IHC 0 excluded) breast cancer	Five treatment arms - (durvalumab + paclitaxel) **vs** - (durvalumab +paclitaxel + capivasertib) **vs** - (durvalumab + paclitaxel + oleclumab) **vs** -(durvalumab + T-DXd) **vs** -(durvalumab + Dato-DXd)	Active, not recruiting		cORR: 100% (4/4; only four pts have completed two on-treatment assessments; remaining seven pts still on treatment)
Sacituzumab govitecan versus physician's choice ChT	TROP2	TROPiCS-02[171,172] *NCT03901339*	III	HR + , HER2-advanced BC, heavily pre-treated, ET resistant	543 post hoc analysis: HER2-Low: 283, IHC 0:(217)	(Active, not recruiting		SG versus physician's choice: HER2-Low patients, ORR: 26% versus 12% mPFS: 6.4 versus 4.2 months. IHC 0 patients, ORR: 16% versus 15%, mPFS: 5.0 versus 3.4 months ILD: 0% versus 1% (full population)

Table 1 (continued) | Overview of the clinical trials on ADC in HER2-Low breast cancer

Intervention	Target	Study name	Phase	Patient population	Number	Trial status	Key goal	Key results (or estimated primary completion date)
Sacituzumab govitecan versus physician's choice	TROP2	ASCENT[139,173] *NCT02574455*	III	TNBC (local advanced or mBC) RR to ⩾2 previous ChT regimens	468 (post hoc analysis: HER2-Low: 123, IHC 0: 293)	Completed	SG versus physician's choice (post hoc analyses): HER2-Low patients, ORR: 32% versus 8%, mPFS: 6.2 versus 2.9mo, mOS: 14.0 versus 8.7mo IHC 0 patients, ORR: 31% versus 3%, mPFS: 4.3 versus 1.6mo, mOS: 11.3 versus 5.9mo ILD: n = 1 (SG group; full population)

AEs Adverse events, BC, breast cancer, BM brain metastases, BOR best overall response, cORR confirmed objective (or overall) response rate, ChT chemotherapy, DOR duration of therapy, ET Endocrine therapy, est. estimated, FAS full analysis set, HER2 human epidermal growth factor 2, HR hormone receptor, IHC immunohistochemistry, ILD interstitial lung disease (or pneumonitis), LMC leptomeningeal carcinomatosis, mBC metastatic breast cancer, mo months, mOS median overall survival, mPFS median progression-free survival, N/A not available, ORR overall response rate, pCR pathologic complete response, R/R relapsed or refractory, SG sacituzumab govitecan, TNBC triple-negative breast cancer.

accurate recognition of these HER2 BC groups[15]. The United Kingdom updated HER2 guidelines introduced the term HER2-Low and attempted to refine its definition to assist pathologists in identifying these tumours in the clinical setting[13]

While new quantitative assays, such as the HERmark BC assay[47,48], immunofluorescence-based automated quantitative analysis[40,49], quantitative IHC[50], and assessment of HER2 mRNA expression levels[51] are being evaluated, any new assay or kit designed for the detection of HER2-Low tumours must undergo rigorous validation through clinical trials before being integrated into clinical practice. Moreover, it is essential that these assays demonstrate superiority over existing methods to be deemed clinically relevant. In our view, the most immediate and impactful progress can be achieved by prioritising the standardisation of existing HER2 testing assays. This involves harmonising laboratory protocols to reduce interlaboratory variability and enhance diagnostic accuracy. Strict adherence to established guidelines, particularly those addressing pre-analytical and analytical variables such as tissue fixation, processing, staining techniques, and scoring criteria, is critical for ensuring consistent and reliable HER2 assessment[6,20,39,44,52,53], Equally important, is the need to improve concordance among pathologists in the interpretation and categorisation of HER2 status, especially as the distinction between HER2-Low, Ultralow, and null becomes increasingly clinically relevant. Targeted training programmes and continuing education are essential to enhance diagnostic precision, support consistent application of scoring systems, and minimise subjectivity in borderline cases[43,54]. With the revolution of digital pathology and artificial intelligence (AI), studies have explored the use of AI and digital image analysis to improve concordance among pathologists in scoring HER2-Low BCs. AI algorithms and deep learning models have demonstrated high accuracy in distinguishing between HER2 IHC scores, particularly in the critical differentiation of 0, 1 +, and 2+ categories[55]. This could provide quantitative and reproducible assessments, reducing subjectivity and inter-observer variability, especially in heterogeneous or borderline cases[9,56]. While these technologies are not yet universally adopted, they represent promising tools for standardising HER2-Low assessment and supporting more consistent clinical decision-making.

The correlation between target antigen expression and ADC efficacy

With the increasingly promising and unexpected efficacy of ADCs in tumours expressing very low levels of the target protein, it is worth briefly exploring their mechanism of action in an effort to understand their therapeutic role better.

ADCs consist of a humanised monoclonal antibody (mAb) linked to a cytotoxic agent, known as the payload, via a molecular linker. Typically, the antibody component recognises and binds to its target antigen on the cell surface, triggering internalisation of the entire ADC complex via endocytosis. This process delivers the payload directly inside cancer cells, reducing systemic exposure to the cytotoxic agent. It is also proposed that bystander killing of the tumour cells may occur if the payload is released into the extracellular space, where it can be taken up by neighbouring HER2-negative tumour cells, leading to their death[5,57] (Fig. 3).

Selecting a suitable target antigen is crucial for modulating the specificity and effectiveness of an ADC. Ideally, the target should be exclusively or predominantly expressed at high levels on tumour cells and minimally on normal cells. HER2 and trophoblast cell surface antigen 2 (TROP2) are examples of such targets currently being utilised in ADC development for the treatment of BC[58,59]. In general, the efficacy of ADCs appears to be closely linked to the level of target antigen expression on tumour cells. For instance, a study using HER2-targeted liposomal doxorubicin demonstrated that nuclear delivery of doxorubicin increased progressively with higher HER2 expression levels, with a threshold effect observed at approximately 200,000 HER2 receptors per cell[60]. This threshold supports previous findings that human cardiomyocytes, which express normal levels of HER2 below this threshold, show little to no uptake of the drug. HER2 receptor density on BC cells ranges from 100,000 to 500,000 molecules per cell in HER2 IHC score

Table 2 | Summary of the ongoing clinical trials for management of HER2-expressing BC*

Intervention	Target	Study name	Phase	Patient population	Number	Key endpoint	primary completion date
HER2-Low (early and advanced BC)							
Disitamab vedotin (RC48) vs physician's choice	HER2	Randomised, parallel control, multicentre study NCT04400695	III	HER2-Low mBC, previous use of anthracyclines, and 1–2 systemic ChT regimens	366 (est.)	PFS	December 2024
Disitamab vedotin (RC48)	HER2	Open-label study NCT05331326	II	HER2-positive and HER2-Low mBC with abnormal PAM pathway activation	64 (est.)	PFS ORR	August 2024
Disitamab vedotin (RC48) + penpulimab	HER2	Open-label study NCT05726175	II	Neoadjuvant treatment of HER2-Low early or locally advanced breast cancer	20 (est.)	pCR	August 2024
Sacituzumab govitecan versus physician's choice	TROP2	SASCIA NCT04595565	III	HER2 IHC score 0–1+ or FISH negative, post-neoadjuvant residual disease	1332 (ER+ and ER-)	iDFS	March 30, 2027
Disitamab vedotin (RC48-ADC) + ET	HER2	Rosy NCT05904964	III	1st line endocrine resistant HER2-Low mBC	288(est.)	PFS	March 2028
T-DXd + /− Anastrozole	HER2	TALENT Randomised, open-label, dual-arm NCT04553770	II	Neoadjuvant Early-stage BC HER2-Low HR+	88	pCR	30-Sep-25
T-DXd + pembrolizumab	HER2	Two-part, open-label Study NCT04042701	I	HER2-Low, advanced/ mBC with failure on prior standard therapy	105	pCR	2025
T-DXd	HER2	HER2-PREDICT Multicentre, observational study NCT04257162		Using biological samples from patients treated with DS-8201a in the metastatic setting	180	Optimal ERBB2 mRNA cut-point predictive of T-DXd response	December 2024
HER2-positive (early and advanced BC)							
T-DXd monotherapy or T-DXd followed by taxane, trastuzumab and pertuzumab vs ddAC and THP	ADC (HER2)	DESTINY-Breast11 NCT05113251	III	HER2-positive, T0-4, N1-3, M0 or ≥T3, N0, M0	927	pCR	December 20, 2024
T-DXd vs T-DM1	ADC (HER2)	DESTINY-Breast05 NCT04622319	III	HER2-positive, post-neoadjuvant residual disease (T1N0 tumours not eligible)	1,660	iDFS	December 31, 2025
T-DM1 + atezolizumab T-DM1 + placebo	ADC (HER2) + ICI (PD-L1)	ASTEFANIA NCT04873362	III	HER2 + +, post-neoadjuvant residual disease (T1mi/T1a/T1b/N0 not eligible)	1700	iDFS	June 25, 2026
Taxane, trastuzumab, pertuzumab, surgery. If pCR: Trastuzumab pertuzumab; if RD, T-DM1	**Monoclonal antibody (HER2)**	**CompassHER2-pCR NCT04266249**	**II**	**Single arm study; HER2-positive, if cN0 T must be >2.0 cm (T2-3); if cN1-2, must be T1-3. T4 or N3 not eligible**	**2156**	**RFS**	**October 31, 2026**
T-DM1 + tucatinib vs T-DM1 + placebo	ADC (HER2	CompassHER2 RD NCT04457596	III	HER2-positive, T1-4, N0-3, post-neoadjuvant residual disease (if ER + , N1 residual disease)	1,031	iDFS	January 2028
T-DXd + pertuzumab; T-DXd + placebo vs Taxane, trastuzumab, and pertuzumab	ADC (HER2),	DESTINY-Breast09 NCT04784715	III	HER2-positive, no prior ChT, no prior HER2-targeted tx for mBC; if neoadjuvant ChT or HER2-targeted tx, DFI > 6 mos	1157	PFS	March 28, 2025
T-DXd single arm study	HER2	DESTINY-Breast12 NCT04739761	III	HER2-positive patients with or without brain metastasis who have a previously treated advanced or metastatic HER2-positive BC	506	ORR in pts without BM PFS in pts with BM	February 2026-
T-DXd combinations (durvalumab, pertuzumab, paclitaxel, tucatinib)	HER2	DESTINY-Breast07 NCT04538742	Ib/II	Safety, tolerability, and anti-tumour activity of T-DXd in combination with other anti-cancer agents in patients with HER2-positive mBC	245	AEs	January 2025
	TKI (HER2)		III		650	PFS	October 31, 2024

Table 2 (continued) | Summary of the ongoing clinical trials for management of HER2-expressing BC*

Intervention	Target	Study name	Phase	Patient population	Number	Key endpoint	primary completion date
Tucatinib, Trastuzumab, Pertuzumab (maintenance) vs Placebo, trastuzumab, Pertuzumab (maintenance)		HER2CLIMB-05 NCT05132582		HER2-positive, untreated mBC, prior 4–8 induction cycles with trastuzumab, pertuzumab, and taxane			
RC48-ADC vs Lapatinib + Capecitabine	HER2	NCT03500380	II/III	Patients with HER2-positive mBC with or without liver metastases	301	PFS	December 2024
Giredestrant + Pertuzumab, trastuzumab (maintenance) vs Pertuzumab trastuzumab (maintenance)	SERD (ER)	heredERA NCT05296798	III	ER + /HER2-positive, ≤1 prior ET for mBC, prior 4–6 induction cycles with trastuzumab-pertuzumab and taxane; if neoadj non–ET-based therapy, DFI > 6 mos	812	PFS	August 31, 2026

*Source is clinicalTrial.gov, accessed on 5th of August 2024.

ADC Antibody drug conjugates, AEs Adverse events, BC breast cancer, BM brain metastases, ChT chemotherapy, ddAC dose-dense Adriamycin (doxorubicin) and Cyclophosphamide, ER Oestrogen receptor, ET Endocrine therapy, est. estimated, FISH Fluorescent in situ hybridisation, HER2 human epidermal growth factor 2, HR hormone receptor, ICI immune check point inhibitor, DFI Disease free interval, iDFS invasive disease free survival, IHC immunohistochemistry, mBC metastatic breast cancer, mo months, ORR overall response rate, pCR pathologic complete response, PFS progression-free survival, PAM PI3K/Akt/mTOR, RD residual disease, SERD Selective oestrogen receptor degrader, T-DM1 Trastzumab ematensine, T-DXd trastuzumab deruxtecan, THP taxanes, trastuzumab, pertuzumab, TKI Tyrosine kinase inhibitor, tx therapy.

1+ and 2+ tumours, compared to over two million in HER2 IHC score 3+ cells[61].

Disitamab vedotin (RC48-ADC) is another ADC composed of a novel anti-HER2 humanised antibody, hertuzumab, conjugated to a microtubule inhibitor monomethyl auristatin E (MMAE) payload through a cleavable linker. This compound showed effective therapeutic activity in a cohort of 48 patients with HER2-Low mBC, reporting an overall response rate (ORR) of 40% and a median PFS of 5.7 months. However, the response was higher in the HER2 IHC score 2 + /ISH-negative subgroup compared with that in patients with HER2 IHC 1+ tumours[62]. Furthermore, results from the phase II DAISY trial, that reported a longer PFS in the HER2 overexpressing cohort and a shorter PFS in the HER2 non-expressing cohort (adjusted HR: 1.96, 95% CI 1.21–3.15, $P = 0.006$) compared to the HER2-Low cohort, in patients receiving T-DXd therapy for mBC[38], supports a positive correlation between HER2 antigen levels and the efficacy of HER2-targeting ADC agents.

In the pre-clinical setting, a study which assessed the in vitro potency of anti-HER2 ADC in a panel of cancer cell lines with various degrees of HER2 protein expression also demonstrated that the extent of internalisation was largely influenced by the cell surface HER2 level on the cell surface, which reduced as the HER2 receptor density decreased[63].

Bystander killing effect in non-target expressing tumours

The degree to which an ADC mediates bystander killing depends on various factors, including the extent of ADC internalisation after binding to the target antigen, the presence of a non-cleavable or cleavable linker, and the hydrophobicity of the attached cytotoxic payload. Non-cleavable linker ADCs, e.g. T-DM1, are primarily effective against the target antigen-expressing cell after internalisation and are most effective in the treatment of cancers that have high and homogenous expression of the target antigen[64–66]. T-DXd incorporates a cleavable linker, which can be cleaved at a defined pH range or by specific proteases to release the cytotoxic agent. This may result in direct target cell death but may also lead to bystander killing depending on the physicochemical properties of the cytotoxic agent[64,67,68]. Consequently, these ADCs are useful for treating tumours with heterogeneous antigen expression[64,69,70].

The translational analysis performed in the DAISY trial demonstrated that the therapeutic impact of cytotoxic agents on HER2-expressing tumour cells may be diminished by the presence of surrounding target-negative tumour cells[71] (Fig. 4). DAISYII trial also explored T-DXd distribution during treatment through IHC staining using an Ac anti-DXd H score, and demonstrated that tumour cells with a high level of HER2 expression presented strong T-DXd levels or staining.

However, two of three patients with BC classified as HER2 IHC 0 who had no or low levels of T-DXd had a confirmed partial response[38]. These data, together with the results of T-DXd in HER2-Ultralow tumours as proven in the DESTINY-Breast-06 trial[37,38], suggests that the effect of T-DXd may be partly mediated by HER2-independent mechanisms and question the need to identify very low levels of HER2 expression as there is no evidence so far to confirm the lack of a similar effect in HER2 null BC.

Mechanisms of resistance to ADC therapy

Understanding the mechanisms behind resistance to ADCs remains a significant challenge. A multicentre study found that cross-resistance may depend on both the antibody target and the cytotoxic payload used in subsequent treatments. Tumour sequencing identified mutations in topoisomerase-related pathways that may contribute to resistance, highlighting the importance of personalised genomic profiling and further research to optimise ADC sequencing for improved patient outcomes[72–74].

A common mechanism of resistance is reduced target antigen expression, which hinders effective antigen–antibody binding[75]. In triple-negative BC (TNBC), clinical resistance to IMMU-132 was observed following the loss of TROP2 expression[76,77]. Similar findings have been reported in other malignancies: CD30 downregulation led to brentuximab vedotin resistance in Hodgkin lymphoma[43,78], and reduced CD33 expression

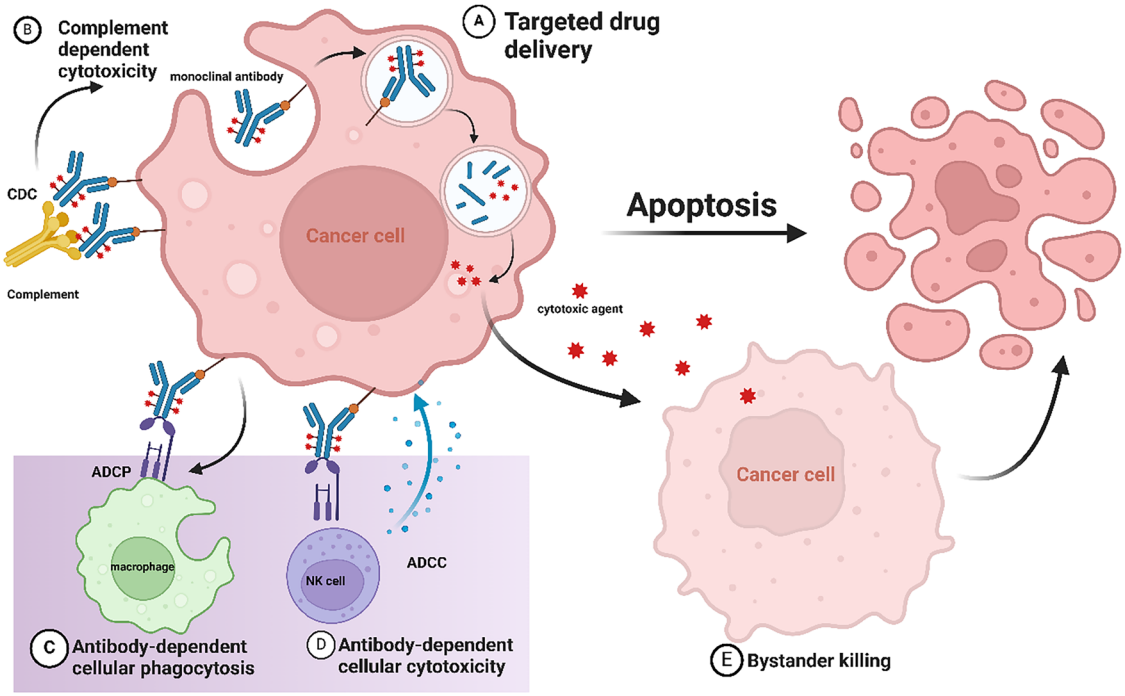

Fig. 3 | Schematic illustration summarising the mechanism of action of the antibody drug conjugate. NK Natural killer. Created by BioRender.com.

was associated with poorer outcomes in acute myeloid leukaemia treated with gemtuzumab ozogamicin[79]. This phenomenon has also been observed with HER2-targeting ADCs such as T-DM1. A phase II trial showed that higher HER2 expression correlated with greater T-DM1 efficacy[80], while separate studies found reduced HER2 levels and binding in resistant cell lines[81,82].

The tumour microenvironment (TME) also contributes to resistance. Cancer-associated fibroblasts (CAFs) can create physical barriers to ADC penetration, and tumour-associated macrophages (TAMs) may clear ADCs via Fcγ receptor-mediated uptake[83–85]. Additional mechanisms include altered intracellular trafficking, impaired internalisation or recycling of the ADC, and tumour heterogeneity, which allows subclones with low or absent antigen expression to evade targeting[85,86].

Resistance related to the payload itself involves drug efflux pumps such as MDR1, mutations in payload targets (e.g., tubulin or topoisomerase), and dysfunctional lysosomal processing, which limits effective payload release[75,84,86].

HER2-positive BC and response to therapy: protein overexpression or gene amplification?

In HER2 overexpressing BC, the growth of tumour cells and hence the response to HER2 pathway targeting therapy is mainly driven by the oncogenic activity of the HER2 signalling pathways. To date, patients with BC featuring either HER2 protein overexpression (IHC 3 +) or equivocal protein expression (IHC 2 +) with evidence of *HER2* gene amplification on ISH studies are considered candidates for anti-HER2 targeted therapies[6]. However, the response rate in patients with HER2-positive BC is not uniform. The magnitude of response, measured by the rate of pathologic complete response (pCR), of HER2 IHC score 3+ tumours ranges from 55 to 70% compared to 17–20% in HER2 2 + /ISH-positive[14,87–91], raising the possibility that HER2 protein overexpression rather than *HER2* gene amplification is the key driver of response (Table 3). Despite the previous data that showed a strong correlation between HER2 gene amplification and response to treatment, most BCs with high-level amplifications also showed IHC 3+ expression, which may have biased the results. Detailed analysis of IHC 2 + /ISH positive BC response to anti-HER2 therapy in the adjuvant setting was based on subgroup analysis with demonstration of some response in patients with HER2 amplified compared to those with non-amplified BC based on the definition of gene amplification rather than selecting a cutoffs that defined a significant difference in response.

The fact that most anti-HER2 targeted therapies exert their action on the HER2 protein and the excellent correlation between HER2 IHC 3+ and high *HER2* gene amplification levels[17,92] may explain why most BCs with high amplification levels respond well to anti-HER2 therapy. Variation in response to treatment is more often seen in HER2 IHC 2+ tumours which frequently show borderline gene amplification[88,93,94]. These observations together with the finding that 4% of BCs with HER2 IHC scores 0 and 1+ show evidence of gene amplification[93], argue against using ISH alone as a predictor of response to anti-HER2 therapy. The HERA and N9831 trials concluded that *HER2/CEP17* ratio and *HER2* gene copy number were not associated with patient outcome[95–97], consistent with the results of several previous studies, which demonstrated that HER2 protein overexpression is the strong predictor of response to anti-HER2 therapy[88,96,98–104]. Other studies have also indicated that the rate of pCR in the subset of patients with evidence of *HER2* gene amplification, in the absence of HER2 protein overexpression, was significantly lower (17% vs 66%)[88,100].

Some studies have suggested a correlation between *HER2* gene amplification levels and the therapeutic response to anti-HER2 therapy[105–108]. However, research by Gianni et al.[109], Xu et al.[110], Guiu et al[108]. and Perez et al.[96,108–110] found no link between *HER2* gene copy number and survival. While some authors, including Zabaglo et al.[111], argued that HER2 protein expression levels do not influence clinical management with anti-HER2 therapy, their study assessed a spectrum of HER2 staining rather than applying a dichotomised classification (IHC 3+ versus 2 +). Other reports have suggested that the optimum strategy for selecting patients for anti-HER2 therapy is to measure *HER2* gene amplification by ISH testing[112,113]. This was based on studies that included tumours with IHC scores of 3+ and 2+ and the ISH negative group included only IHC 2+ tumours, resulting in an over-emphasis of the value of ISH testing[112,113]. A recent study recommended performing FISH testing in HER2 IHC-negative cases (scores of 0 and 1 +), as some of these tumours were found to be FISH positive albeit near the *HER2/CEP17* ratio and HER2 gene copy number cut-offs[114]. We argue that their conclusion is debatable, as it was not supported by evidence of therapeutic response. Additionally, in

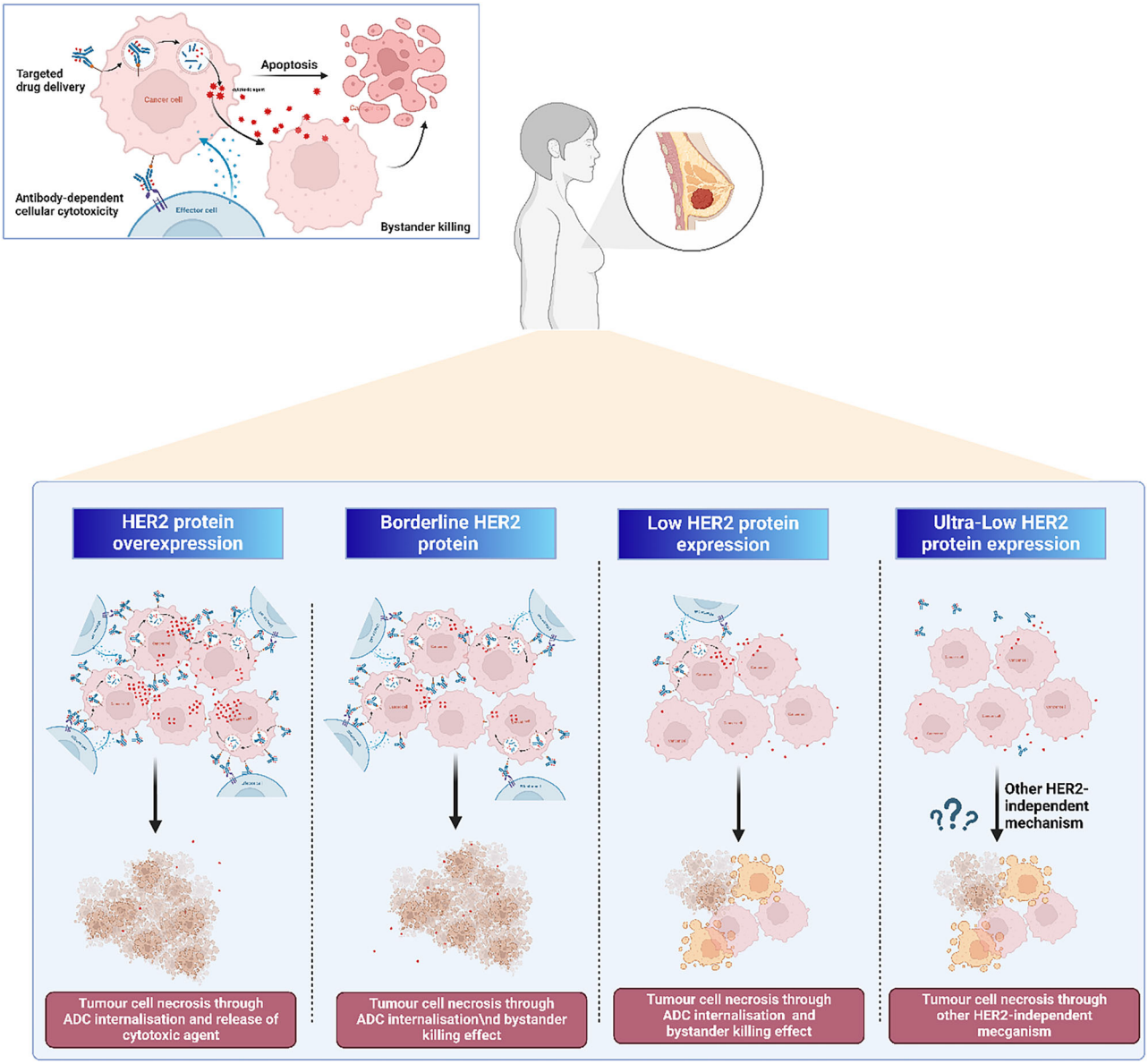

Fig. 4 | Schematic illustration describing the difference in the mechanism of action of the antibody drug conjugate in different levels of HER2 expression, with the role of the bystander killing effect. Created by Biorender.com.

light of the promising outcomes of ADC in HER2-Low cases, cost-effective analysis of conducting FISH testing for every HER2-negative patient to qualify for anti-HER2 therapy, as opposed to directly administering ADC is a consideration.

The phenomenon of oncogenic addiction of cancer cells to HER2 for the maintenance of their malignant phenotype is one of the main determining factors of response to anti-HER2 therapies[115,116]. High HER2 protein expression, *HER2* mRNA level, high and homogenous pattern of *HER2* gene amplification, and HER2 downstream signalling are necessary for true HER2 addiction, and identification of HER2-addicted tumours is essential to reap the complete therapeutic benefit of HER2-targeted drugs[117,118]. Furthermore, the HER2-enriched (HER2-E) molecular subtype identifies patients with increased likelihood of achieving a pCR following neoadjuvant anti-HER2-based therapy[119–123]. Furthermore, the PAMELA phase 2 trial concluded that the HER2-E subtype is the key predictor of pCR following trastuzumab and lapatinib without chemotherapy in early-stage HER2-positive BC[124].

In an effort to further investigate the differential response of both HER2-positive BC classes to HER2 targeted therapy, our group investigated the pCR rates, patient outcome and the differential expression of HER2 oncogenic signalling genes in a large series of patients ($n = 7390$) with BC, classified as HER2-positive due to HER2 protein overexpression or borderline expression but with *HER2* gene amplification[14]. The results of our study suggest that BC characterised by HER2 protein overexpression (IHC 3 +) is likely propelled by the HER2 oncogenic signalling pathway, alongside increased expression of several receptor tyrosine kinases (RTKs) e.g. FGFR4, EGFR, and HER2 itself. Additionally, these tumours exhibit elevated expression of genes within the HER2 amplified region on Chr17q12-q21, including GRB7, which may account for their heightened sensitivity to anti-HER2 therapy. Conversely, patients with borderline HER2 protein expression show limited response to anti-HER2 therapy, irrespective of HER2 gene amplification levels, particularly those with oestrogen receptor-positive (ER-positive) tumours. Such low response rates of BC with borderline HER2 protein expression, which appear not to differ from the response rates of HER2-negative BC to chemotherapy (15–25%)[125], may support the need for further comparative studies between conventional anti-HER2 therapy and HER2 directed ADCs in patients with these tumours.

Table 3 | Overview of previous studies that investigated the differential response of HER2-positive classes to therapy

Neoadjuvant Setting	Study Title	Rate of pathologic complete response			*P* value
		Overall	HER2 IHC 3 +	HER2 IHC 2 + /ISH positive	
Yan et al.[13]	Association Between the HER2 Protein Expression Level and the Efficacy of Neoadjuvant Chemotherapy in HER2-Positive Breast Cancer	37.2% (110/296)	41.4% (99/139)	19.3% (11/75)	0.003
Krystel-Whittemore et al.[14]	Pathologic complete response rate according to HER2 detection methods in HER2-positive breast cancer treated with neoadjuvant systemic therapy	59% (330/560)	67% (303/455)	17% (12/76)	<0.001
Rakha et al.[15]	Retrospective observational study of HER2 immunohistochemistry in borderline breast cancer patients undergoing neoadjuvant therapy, with an emphasis on Group 2 (HER2/CEP17 ratio≥ 2.0, HER2 copy number< 4.0 signals/cell) cases	25% (184/749)	54% (79/146)	19% (35/180)	<0.001
Xu et al.[16]	HER2 protein expression level is positively associated with the efficacy of neoadjuvant systemic therapy in HER2-positive breast cancer	43.7% (73/167)	47.9% (67/140)	22.2% (6/27)	0.014
Zhao et al.[17]	HER2 immunohistochemistry staining positivity is strongly predictive of tumour response to neoadjuvant chemotherapy in HER2-positive breast cancer	64.5% (49/76)	80% (40/49)	34% (9/40)	<0.001
Chen et al.[18]	Pathologic complete response to neoadjuvant anti-HER2 therapy is associated with HER2 immunohistochemistry score in HER2-positive early breast cancer	74/ 181 (40.9%)	46%	25%	0.016
Atallah et al.[19]	Differential response of HER2-positive breast cancer to anti-HER2 therapy based on HER2 protein expression level	26% (194/751)	57% (88/155)	22% (35/162)	0.001
Chen et al.[20]	Differences between the efficacy of HER2(2 +) /FISH-positive and HER2 (3 +) in breast cancer during dual-target neoadjuvant therapy	59.8%	63.3%	41.7%	<0.05
Jiao et al.[21]	Comparison of the Response to Neoadjuvant Therapy Between Immunohistochemistry HER2 (3 +) and HER2 (2 +)/ISH+ Early-Stage Breast Cancer: A Retrospective Multicentre Cohort Study	35.5%	40%	18%	<0.001
Winer, Ruth et al.[22]	Immunohistochemical Status Predicts Pathologic Complete Response to Neoadjuvant Therapy in HER2-Overexpressing Breast Cancers	49%	54%	22%	<0.001

CEP chromosome enumeration probe, HER2 Human epidermal growth factor receptor 2, IHC Immunohistochemistry, ISH in situ hybridisation.

Role of ER expression in theresponse of HER2-positive BC classes to therapy

The interaction between HER2 and ER signalling pathways plays an important role in driving BC proliferation, involving intricate molecular processes[126]. It has been reported that HER2 overexpression may reduce the effectiveness of anti-endocrine therapy (tamoxifen and aromatase inhibitors (AI)) and ovarian suppression in pre-menopausal women[127–134]. Regarding resistance to anti-HER2 therapy, in the pre-clinical setting, it has been shown that the expression of ER and its downstream targets is increased in cells with acquired resistance to anti-HER2 therapy[130,135]. Reactivation of ER expression and signalling, including a switch from ER-negative to ER-positive status, has been observed in HER2-positive tumours following neoadjuvant lapatinib treatment[130].

The interplay between ER and HER2 levels in predicting response to adjuvant trastuzumab within HER2-positive classes was considered in the secondary analyses of HERA and NeoALTTO pivotal trials. Notably, patients with ER-positive and HER2-positive status, but with low HER2 gene copy number, derived diminished benefit from adjuvant trastuzumab despite receiving concomitant endocrine therapy[136]. Additionally, investigations into gene expression levels of ESR1 and HER2 revealed that higher HER2 and lower ESR1 levels correlated with increased pCR rates for HER2-targeted therapy[137]

Despite extensive research, the response variability to anti-HER2 therapy among HER2-positive BC classes remains complex across distinct ER expression groups, necessitating further investigation. In an earlier study, we investigated the role of ER positivity among HER2-positive categories of BC and its role in response to anti-HER2 therapy. We concluded that ER positivity was a significant predictor of poor response to anti-HER2 therapy in the overall cohort and tumours with borderline protein expression but not in patients with HER2 IHC 3 + BC, observed in both neoadjuvant and adjuvant treatment settings[14]. These findings resonate with those of Harbeck, Gluz[138] who reported comparable pCR rates in HR-positive and HR-negative patients with HER2 protein overexpression. This is likely because tumours with HER2 protein overexpression are primarily HER2-E, dominantly driven by the HER2 signalling pathway regardless of ER level, with blockade of HER2 signalling leading to apoptosis and tumour cell death. In contrast, HER2-positive tumours with borderline HER2 protein expression are more frequently ER-positive, enriched with ER signalling pathways and associated genes, e.g. ESR1 and BCL2, dependent on the alternative ER signalling to survive, and less likely to include tumours with the HER2-E molecular subtype[14].

Optimal ADC sequence and patient selection

Considering the expansion of the therapeutic armamentarium for BC patients, with several ADCs approaching the clinic, further research is needed to master treatment sequencing in mBC. In post hoc analyses of the phase III TROPiCS-02 and the ASCENT trials[139,140], the anti-TROP2 ADC, Sacituzumab govitecan (SG) demonstrated efficacy in HER2-Low mBC, both HR-positive and HR-negative tumours. SG has been approved after disease progression on ET and ⩾2 additional systemic therapies for mBC[139,140]. Another TROP2-directed ADC, Dato-DXd, a topoisomerase I inhibitor, has shown efficacy following prior ET and 1–2 lines of chemotherapy in the TROPION-Breast01 trial[141]. In this study, median mPFS increased from 4.9 months with the treatment of the physician's choice to 6.9 months with Dato-DXd (HR, 0.63 [95% CI, 0.52 to 0.76][141] An overview of the clinical trials on ADC in HER2-Low BC is summarised in Table 2.

HR-positive/HER2-Low

For patients with HR-positive/HER2-Low mBC, the standard first-line therapy remains a combination of ET and CDK4/6 inhibitors (CDK4/6i) (abemaciclib, palbociclib, or ribociclib), provided there is no visceral crisis[142,143] (Fig. 5).

For subsequent lines of therapy, additional ETs, possibly combined with mTOR or PI3K inhibitors, can be considered[144]. After ET regimes are

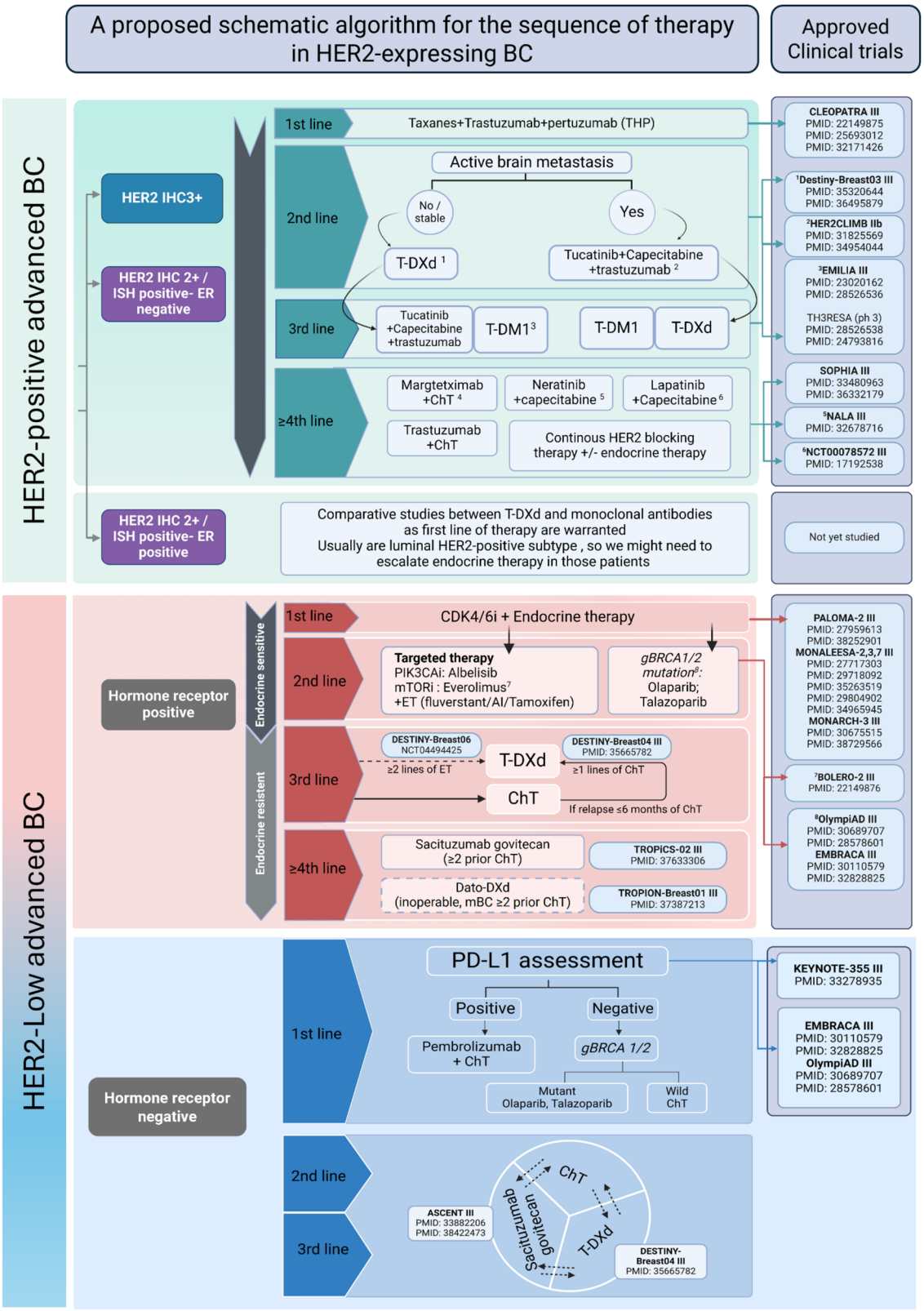

Fig. 5 | A schematic illustration presenting a proposed sequence of therapy in HER2-expressing tumours. BC breast cancer, ChT chemotherapy, ER oestrogen receptor, ET Endocrine therapy, ISH in situ hybridisation, HER2 human epidermal growth factor 2, HR hormone receptor, IHC immunohistochemistry, mBC metastatic breast cancer, T-DM1 trastuzumab emtansine, T-DXd trastuzumab deruxtecan, THP taxanes, trastuzumab, pertuzumab. Created with biorender.com.

exhausted, single-agent chemotherapy is recommended[142,143]. However, with the accumulating evidence of favourable results with T-DXd over chemotherapy in patients who received ⩾2 ET in the DESTINY-Breast06 trial[37] and following prior ET and up to two lines of chemotherapy in the DESTINY-Breast04 trial[15], we believe that it may be superior to standard chemotherapy, potentially positioning T-DXd as the first non-ET for this population. Considering SG, although it showed significant improvements in PFS (5.5 vs. 4.0 months; HR: 0.66) and overall survival (14.4 vs. 11.2 months; HR: 0.79) compared to chemotherapy, patients in the TROPiCS-02 trial received considerably more prior treatment (ET, CDK4/6 inhibitors, and 2–4 lines of chemotherapy) than those in the DESTINY-Breast04 trial, rendering it more suitable as a later treatment option, following T-DXd.

HR-negative/HER2-Low

The DESTINY-Breast04 trial positions T-DXd as a second-line treatment after chemotherapy, or as a frontline option for patients experiencing early recurrence after neo-adjuvant chemotherapy. Although the trial was not specifically powered to evaluate TNBC, T-DXd showed a clear benefit in the small TNBC subgroup, suggesting its use in patients with TNBC with low HER2 expression[15]. SG is also approved for patients with TNBC who have received at least two prior systemic therapies based on the ASCENT trial[139], which demonstrated a significant PFS and overall survival benefit over chemotherapy.

For patients eligible to receive both TROP-2- and HER2-targeted ADCs, it is still uncertain which agent should be used first and whether administering them in sequence provides added clinical benefit. In a real-world cohort of heavily pretreated patients with HER2-Low disease, SG retains its clinical activity. T-DXd continues to demonstrate promising clinical activity after SG, supporting the use of sequential ADC in this population[43,145]. In a multicentre analysis[72–74], researchers investigated the use of sequential ADCs in patients with HR + /HER2-Low and HR-/HER2-Low mBC. Across all subgroups, the overall response rate (ORR) and time to next treatment (TTNT) were higher following the first ADC than for the second, regardless of HR status or the order of ADC administration. Additionally, in this heavily pre-treated population, PFS was shorter with the second ADC, highlighting a potential limitation of sequential ADC use.

According to DESTINY-Breast 04 trial, patients were enroled in the T-DXd arm after receiving one to two lines of chemotherapy, while in the SG based trials, patients received two to four lines of chemotherapy prior to enrolment in the SG arm. Regarding HR-status, both T-DXd and SG showed promising results in HR-positive and HR-negative BC patients, although the DESTINY-Breast 04 included only 63 patients with HR-negative, HER2-Low disease.

Until there is a randomised-controlled clinical trial that elucidates the comparative efficacy of these agents, current evidence supports the use of T-DXd over SG in the management of HER2-Low BC patients who meet the eligibility criteria in view of the higher level of evidence in the HER2-Low population (Phase III versus post hoc analyses for SG), in addition to the lower number of prior lines of chemotherapy received by patients in DESTINY-Breast04.

Identifying patients most likely to benefit from these therapies may require more detailed tumour biomarker assessment, along with consideration of individual risk profiles for treatment-related toxicity. As part of the selection process, it is important to evaluate tolerance to potential side effects. Common adverse events include neutropenia, anaemia, fatigue, nausea, and diarrhoea. More serious, though less frequent, toxicities such as interstitial lung disease (particularly with T-DXd), cardiac dysfunction, and ocular complications should also be taken into account[42,146–149].

Potential value of ADCs in HER2 IHC 2+ and ISH-positive BC. The Phase III KATHERINE trial, comparing adjuvant T-DM1 versus trastuzumab for residual invasive disease following neoadjuvant therapy for HER2-positive BC, reported that treatment benefit was less pronounced in patients with HER2 IHC 2 + /ISH-positive compared to that observed in patients with IHC 3+ tumours in the trastuzumab arm. However, in the T-DM1 arm, the 3-year invasive disease-free survival (IDFS) rate did not differ significantly between the two HER2-positive groups (89% in IHC 3+ and 85% in IHC 2 + /ISH-positive). Further analysis revealed that patients with heterogeneous HER2 expression, predominantly observed in HER2 IHC 2 + /ISH-positive tumours, had comparable IDFS rates to those with homogeneous expression, mainly observed in HER2 IHC 3 + , in the T-DM1 arm (89% and 88%, respectively). In contrast, less benefit was observed in tumours with heterogeneous HER2 expression in the Trastuzumab arm[150].

In vitro studies that compared trastuzumab and T-DM1 in HER2-positive cell lines revealed that T-DM1 was more efficacious in trastuzumab-sensitive and in trastuzumab-insensitive HER2-positive cell lines. Using a trastuzumab-resistant xenograft tumour model, it was also demonstrated that T-DM1 can induce both apoptosis and mitosis in vivo[151,152]. Additionally, comparative trials are required to explore the potential value of T-DXd therapy versus trastuzumab in tumours with borderline HER2 expression, particularly in view of the effectiveness of this form of treatment in both IHC HER2-positive and HER2-Low BC (Fig. 5).

Novel targets in the management of HER2-positive/ER-positive BC

Approximately 50% of HER2-positive tumours will co-express hormone receptors, which results in substantial clinical heterogeneity in the behaviour of HER2-positive BC[153]. Treating HER2-positive/ER-positive BC is complex. While identification of the biological driver (i.e., whether ER or HER2 signalling is dominant) may assist decision making regarding the choice of therapy, pathway interaction and crosstalk may modify and alter the disease course during treatment. Combining hormone therapy with anti-HER2 agents has proven beneficial in some HER2-positive patients[153,154], particularly those who have tumours with high ER expression. Some ER-positive/HER2-positive tumours pursue a clinical course similar to luminal A tumours (i.e., ER-driven cancer), others behave as HER2-E tumours (HER2-driven cancer), and some show overlapping clinical features that require combined targeted blockade of both ER and HER2 pathways. Patients with luminal HER2-positive disease may benefit from the addition of ET to their neoadjuvant treatment to improve pCR rates (Table 4).

CDK4/6 inhibitors

The cyclin D1-CDK4/6 pathway has been shown to mediate treatment resistance in HER2-positive BC[155–158]. Several clinical trials are evaluating the utility of CDK4/6 inhibitors in addition to ET in HER2-positive BC and the potential scope for a non-chemotherapy approach (PATINA NCT02947685; PATRICIA NCT02448420; MonarcHER NCT02675231; PALTAN NCT02907918). The phase II MonarcHER trial was the first randomised study utising a CDK4/6 inhibitor to report results in combination with ET and anti-HER2 treatment[159]. This study randomly assigned 237 patients with HER2-positive and HR-positive advanced BC who had been previously treated with at least two prior HER2 targeted therapies to abemaciclib, trastuzumab and Fulvestrant (arm A), abemaciclib with trastuzumab (arm b) or chemotherapy plus trastuzumab (arm c). The triple combination was superior to chemotherapy and trastuzumab (8.3 months v 5.7; HRa, 0.67; $P = 0.051$). No difference was observed between abemaciclib and trastuzumab versus chemotherapy and trastuzumab (HRa, 0.94; $P = 0.77$). Further studies are warranted to delineate the role of CDK4/6 inhibitors in HER2-positive and HR-positive BC. In this direction, the PATRICIA phase II trial enroled 71 patients with HER2-positive disease who had received 2–4 prior lines of anti-HER2–based regimens to receive palbociclib plus trastuzumab with or without letrozole (if HR-positive)[160]. The study concluded that the benefit of palbociclib and trastuzumab was mostly restricted to patients with HR-positive disease. More importantly, the luminal subtype, defined by PAM50, was independently associated with improved PFS compared with the non-luminal subtypes (10.6 months vs 4.2 months; adjusted hazard ratio, 0.40; $P = 0.003$)[160]. Based on these results,

Table 4 | Summary of different treatment options for hormone receptor-positive / HER2-positive breast cancer

NCT identifier	Phase	Treatment arm	Cohort (Number of participants)	mPFS
TAnDEM	III	Trastuzumab + anastrozole **versus** anastrozole	HR + /HER2-positive (207)	4.8 versus 2.4 ($p = 0.0016$)
EGF30008	III	Lapatinib + letrozole **versus** letrozole	HR + /HER2-positive (219)	8.2 versus 3.0 ($p = 0.019$)
eLEcTRA	III	Trastuzumab + letrozole **versus** letrozole	HR + /HER2-positive (57)	14.1 versus 3.3 ($p = 0.23$)
PERTAIN	II	Pertuzumab + trastuzumab +AI **versus** trastuzumab + AI	HR + /HER2-positive (258)	18.9 versus 15.8 ($p = 0.007$)
YSUCC-002	III	Trastuzumab + ET **versus** trastuzumab + ChT	HR + /HER2-positive (392)	19.2 versus 14.8 ($p < 0.0001$)
Cyclin dependent kinase 4/6 inhibitor				
NCT02657343	I	Ribociclib + T-DM	HR + /HER2-positive (12)	10.4
MonarcHER	II	Group A: Abemaciclib +trastuzumab + fulvestrant Group B: abemaciclib +trastuzumab Group C: chemotherapy +trastuzumab	HR + /HER2-positive(325)	A versus C: 8.3 versus 5.7 ($p = 0.051$) B versus C: 5.7 versus 5.7 (non-significant)
PATRICIA	II	Palbociclib + trastuzumab	Cohort A: ER− (15) cohort B1: ER + (28) cohort B2: ER+ with letrozole (28)	Luminal A: 10.6, Luminal B: 8.3, HER2-enriched: 4.3, Normal-like tumours: 3.7 **PFS rate at 6 months:** Cohort A versus B1 versus B2: 33.3% versus 42.8%versus 46.4%.
PATINA	III	palbociclib + anti-HER2 therapy (H or HP) + ET	HR + /HER2+ mBC (518)	The addition of palbociclib significantly improved PFS with a HR of 0.74 (95% CI, 0.58–0.94; 1 sided $p = 0.0074$). Median PFS was 44.3 months (95% CI: 32.4–60.9) in the palbociclib arm compared to 29.1 months (95% CI: 23.3–38.6) in the control arm.

AI Aromatase inhibitors, ChT Chemotherapy, ER Oestrogen receptor, ET endocrine therapy, H Herceptin, HER2 human epidermal growth factor 2, HP Herceptin and Pertuzumab, HR hormone receptor, mBC metastatic breast cancer, mPFS median progression-free survival, T-DM1 trastuzumab-ematensine.

the PATRICIA-II phase II trial (NCT02448420) is currently comparing palbociclib, trastuzumab, and endocrine therapy versus chemotherapy and trastuzumab in patients with PAM50 luminal disease. PATINA phase III trial showed that the addition of the CDK4/6 inhibitor palbociclib was of benefit as first-line maintenance therapy in HR-positive, HER2-positive metastatic BC in combination with anti-HER2 and endocrine therapy[161].

Alpha-specific PI3K inhibitors

PIK3CA somatic mutations in HER2-positive disease are frequent and represent approximately 30% of all HER2-positive and HR-positive tumours[162]. Thus, based on the clinical efficacy of alpelisib, an alpha-specific inhibitor, in HER2-negative and HR-positive disease, together with pre-clinical data suggesting a role for PI3K/Akt/mTOR pathway alterations in anti-HER2 treatment resistance, there is a strong rationale for evaluating the combination of alpelisib with anti-HER2 treatment in patients with HER2-positive and HR-positive PIK3CA-mutated BC. A phase I clinical trial of alpelisib and T-DM1 in HER2-positive mBC after taxane-trastuzumab showed that the combination was tolerable and demonstrated activity with an ORR of 43%. Furthermore, activity was also observed in T-DM1–resistant disease (n 510) with an ORR of 30%[163]. These data support the view that PI3K inhibition targets a resistance pathway to anti-HER2 therapy, providing rationale for continued evaluation the role of PI3K inhibition in refractory HER2-positive mBC. Several trials combining anti-HER2 therapy with PI3K pathway inhibition are ongoing, such as the phase I trial IPATHER (NCT04253561) with ipatasertib, the phase I trial B-PRECISE-01 (NCT03767335) with the PI3K inhibitor MEN1611 and the phase III trial EPIK-B2 (NCT04208178) with alpelisib as maintenance therapy.

Conclusion

HER2-Low BC is emerging as a promising targetable clinical entity for ADC therapy with efficacy linked to the level of HER2 expression. The clinical benefit of T-DXd in HER2 IHC 0 tumours is questionable, and whether this BC subset represents a target for T-DXd remains to be defined particularly with initial data showing some response in the HER2-Ultralow category. Standardisation of the current HER2 detection assays, with refinement and adherence to the guideline recommendations regarding pre-analytical and analytical variables for HER2 testing, including scoring criteria, and ongoing pathologist and medical scientist training are important to enhance reproducibility and to improve concordance in categorisation, particularly at the lower end of the HER2 expression spectrum. At the present time, HER2-Low relapsed/mBC is sufficient to warrant consideration of T-DXd treatment for affected patients with an emphasis on optimising therapy sequencing. Further studies comparing conventional anti-HER2 therapy and HER2-directed ADCs in HER2 IHC 2 + /ISH-positive tumours are also warranted to inform clinical practice and improve patient outcome.

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